12TH CHEMISTRY · CHAPTER 28 · SOLVED EXERCISE

Chromatography

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS :

I) Why must the solvent level be below the spot of the dye?

(a) To increase temperature
(b) To prevent the dye from dissolving directly into the solvent
(c) To speed up chromatography
(d) To increase adsorption

Answer: (b) To prevent the dye from dissolving directly into the solvent — Otherwise the spot would wash off into the solvent in the tank instead of moving up the paper.

II) Why do some components of a dye travel further up the paper than others?

(a) They are more soluble in the water trap in paper (stationary phase)
(b) They are more soluble in the solvent (mobile phase)
(c) They react with the paper
(d) They have higher density

Answer: (b) They are more soluble in the solvent (mobile phase) — Components more soluble in the moving solvent are carried further.

III) A substance has a very low R_f value. What does this indicate?

(a) It is insoluble in the solvent
(b) It is highly soluble in the solvent
(c) It strongly sticks to the water inside paper
(d) It has a high density

Answer: (c) It strongly sticks to the water inside paper — It dissolves better in the stationary phase, so it travels only a short distance.

IV) Which statement best explains adsorption?

(a) Substance dissolves completely in another
(b) Substance sticks only to the surface of another material
(c) Substance changes its state
(d) Substance reacts chemically

Answer: (b) Substance sticks only to the surface of another material — Adsorption is a surface phenomenon.

studysapiens.com

V) A polar compound in TLC will usually:

(a) Travel farther up the plate (b) Stay near the baseline
(c) Evaporate quickly (d) Dissolve completely

Answer: (b) Stay near the baseline — It is adsorbed strongly on the polar stationary phase and moves slowly.

VI) Why is TLC considered faster than paper chromatography?

(a) It uses more solvent
(b) It has a solid stationary phase that allows quicker separation
(c) It requires higher temperature
(d) It uses colored substances only

Answer: (b) It has a solid stationary phase that allows quicker separation — The thin solid layer of silica or alumina gives quicker separation.

VII) What is the role of the carrier gas in gas chromatography?

(a) To react with the sample
(b) To act as stationary phase
(c) To carry vaporized sample through the column
(d) To detect the substances

Answer: (c) To carry vaporized sample through the column — The inert gas is the mobile phase.

VIII) Why must the stationary phase in GC have a high boiling point?

(a) To evaporate quickly
(b) To increase pressure
(c) To remain in liquid form during the process
(d) To react with gases

Answer: (c) To remain in liquid form during the process — It must not vaporise in the heated column.

In the textbook the last two options of this MCQ are both lettered "c)"; the fourth is treated as (d).

IX) What does the area under a peak in a chromatogram indicate?

(a) Amount (proportion) of the substance
(b) Retention time
(c) Temperature of column
(d) Color of compound

Answer: (a) Amount (proportion) of the substance — The peak area gives the proportion of each component.

studysapiens.com

X) If two peaks have similar base width but different heights, how can composition be estimated?

(a) By color (b) By retention time

(c) By pressure (d) By peak height

Answer: (d) By peak height — With similar base widths, peak height is used instead of area.

Q2. SHORT ANSWER QUESTION:

a) What is the principle of paper chromatography?

The technique of using a solvent to separate a mixture into its components on paper is called paper chromatography. The separating principle is the **relative solubility** (partition) of the components between two phases: the solvent moving up the paper (mobile phase) and the water combined with the cellulose of the paper (stationary phase).

b) Why is pencil used instead of ink in chromatography to mark the basline?

Pencil (graphite) is insoluble in the solvent, so the line does not move. Ink is a mixture of dyes which would dissolve in the solvent, travel up the paper and mix with the spots of the sample.

c) State two advantages of TLC over paper chromatography.

(1) TLC is faster than paper chromatography. (2) More than one sample can be spotted on the same plate, and a wider choice of locating methods (ninhydrin, ultraviolet light, iodine vapours) can be used on the solid layer.

d) Why do polar substances move slowly in TLC?

Polar substances have a greater attraction for the polar solid stationary phase (silica or alumina) and are adsorbed more strongly. Therefore they travel more slowly up the plate.

e) Why is TLC not suitable for large-scale separation?

Because only small amounts of sample can be applied to the thin layer and analysed.

f) What is the function of injector port in GC?

The sample is injected into the injector port, where it is quickly vaporised by an oven so that the carrier gas can carry its vapours into the column.

g) Why must samples be volatile in GC?

Because the mobile phase is a gas: the sample must be vaporised so that the carrier gas can carry it through the column. Substances which do not vaporise easily cannot be analysed.

h) State two limitations of gas chromatography.

1. GC works only for volatile (easily vaporised) substances.

study-sapiens.com

2. Samples must be thermally stable; compounds that decompose at high temperatures cannot be used.

i) What information does a chromatogram provide?

(1) The number of components in the mixture (number of peaks or spots). (2) The identity of each component, from its retention time (or R_f value). (3) The proportion of each component, from the area (or height) of its peak.

j) What is adsorption? Give one example.

Adsorption is a surface phenomenon in which particles accumulate only on the outer surface of a material. For example, activated charcoal adsorbs gases and impurities onto its surface, which is why it is used in filters and purification systems.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) A mixture of amino acids is analysed using thin layer chromatography. The chromatogram obtained is shown. The table shows some R_f values for different amino acids in the same solvent. [alanine 0.40, glutamic acid 0.29, leucine 0.71, valine 0.61]
I) Use the chromatogram and the R_f values to deduce the amino acid responsible for spot A and spot B.
ii) A second chromatogram of the same mixture is taken using a more polar solvent. Predict the effect on the R_f values of the amino acids. Explain your reasoning.

I) Readings from the printed chromatogram (baseline at 0 cm): solvent front ≈ 5.0 cm, spot A ≈ 3.55 cm, spot B ≈ 1.45 cm.

Rf of A = 3.55 ÷ 5.0 = 0.71 → leucine

Rf of B = 1.45 ÷ 5.0 = 0.29 → glutamic acid

ii) The R_f values will increase. Amino acids are polar, so they are more soluble in a more polar solvent (mobile phase) and are held less by the stationary phase; they therefore travel further up the plate.

3. Gas chromatography involves a stationary phase and a mobile phase.
a) Name a suitable substance that could be used for each phase.

Stationary phase: a non-volatile, long-chain, non-polar hydrocarbon liquid (high boiling point) on silica or alumina particles. Mobile phase: helium or nitrogen (inert carrier gas).

b) A mixture of three organic compounds is separated by gas chromatography. The chromatogram obtained is shown below. [Peaks: ester A = 14, ester B = 25, ester C = 52, ester D = 38]
I) Explain the meaning of the term retention time.
ii) Calculate the % of N in the mixture. Show your working.

I) Retention time is the time taken by a substance to pass through the column.

The question asks for "% of N" and speaks of three compounds, but the printed chromatogram shows four peaks labelled ester A, B, C and D (areas 14, 25, 52 and 38) and no compound N. The percentage of every ester is therefore calculated.

ii) Sum of peak areas = 14 + 25 + 52 + 38 = 129

$$\% \text{ of A} = (14 \div 129) \times 100 = 10.9\ \%$$

$$\% \text{ of B} = (25 \div 129) \times 100 = 19.4\ \%$$

$$\% \text{ of C} = (52 \div 129) \times 100 = 40.3\ \%$$

$$\% \text{ of D} = (38 \div 129) \times 100 = 29.5\ \%$$

c) An unknown amino acid is analysed by thin-layer chromatography. Two chromatographs of the unknown amino acid and four reference amino acids, P, Q, R and S, are obtained using two different solvents. Identify the unknown amino acid. Justify your answer.

The unknown amino acid is **S**.

Solvent	Unknown spot (approx.)	Reference spots at the same height
Solvent 1	1.8 cm	Q and S
Solvent 2	3.8 cm	R and S

Only S travels the same distance (has the same R_f value) as the unknown in **both** solvents. Q matches only in solvent 1 and R only in solvent 2, so they are ruled out.

studysapiens.com

DESCRIPTIVE QUESTIONS

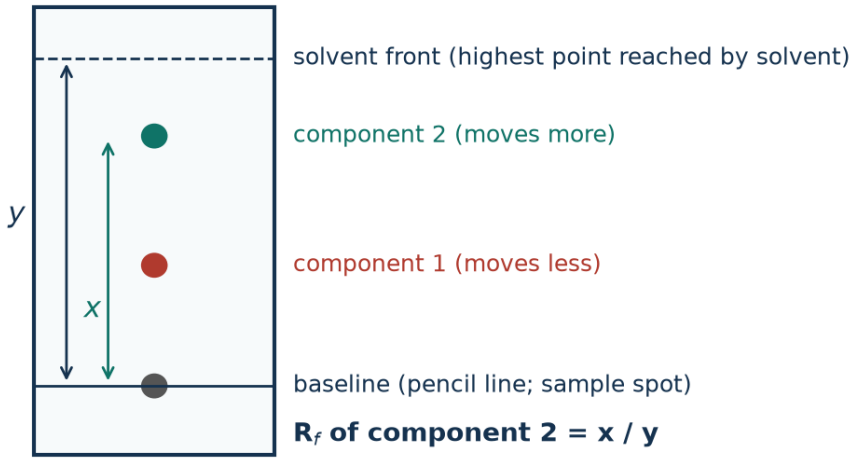
Q4. Describe the principle and working of paper chromatography.

Principle

The technique of using a solvent to separate a mixture into its components on paper is called paper chromatography. The separating principle is the **relative solubility** (partition) of the components between two phases: the solvent moving up the paper (mobile phase) and the water combined with the cellulose of the paper (stationary phase).

Working

1. A small spot of the sample is put on the baseline near the bottom of the paper with a capillary tube.
2. The paper is dipped in a suitable solvent, which soaks up the paper. The solvent level must be below the spot of the sample.
3. The solutes in the sample dissolve in the solvent to different extents. Those more soluble in the solvent (mobile phase) move up the paper with it.
4. Others dissolve better in the water trapped in the paper (stationary phase), so they do not travel very far up the paper.
5. This difference in solubility separates the components. Colourless substances are made visible by spraying a locating agent which gives colour to them.
6. The R_f value of each spot is calculated and compared with known values.



Measuring the distances travelled by a spot and by the solvent front

The R_f value (retardation factor or retention factor) of a substance is calculated by dividing the distance travelled by the component by the distance travelled by the solvent.

Rf = distance travelled by the substance ÷ distance travelled by the solvent

R_f has no unit because it is a ratio. The R_f value of a substance does not change under the same conditions (same solvent and same temperature), so it is used to identify the substance.

Q5. Explain the principle and working process of thin layer chromatography (TLC) in detail.

In thin layer chromatography (TLC), alumina (Al_2O_3) or silica (SiO_2) is mixed with water to make a slurry, which is spread evenly onto a glass slide. This layer is dried in an oven to make a white solid layer (the stationary phase) on the glass slide. TLC is faster than paper chromatography.

Principle (adsorption + solubility): polar substances have a greater attraction for the polar solid stationary phase and are adsorbed more strongly, so they travel more slowly up the plate. Non-polar substances are more soluble in the non-polar solvent and move further up the slide. This difference allows the substances in the sample to separate.

Working

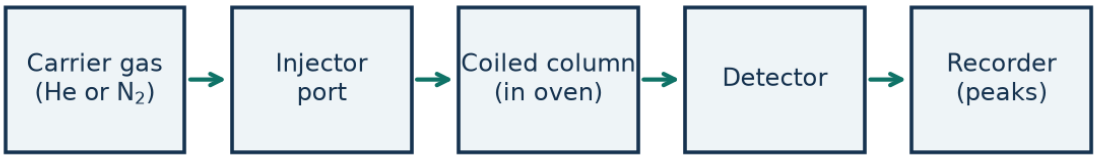
1. A straight baseline is drawn with pencil and ruler near the bottom of the glass slide and a spot of sample is put on it with a capillary tube. More than one sample can be spotted at different places on the same slide.
2. The solvent is taken in a chromatography tank and the slide is dipped in it (mobile phase) so that the baseline is above the solvent level.
3. The components dissolve in the solvent and adsorb onto the aluminium oxide differently: some are more soluble in the mobile phase while some are more adsorbed onto the solid stationary phase.
4. After the solvent has moved up to a certain height, the slide is removed. The solvent front is marked and the slide is dried with a dryer.
5. The R_f values of the spots are calculated and compared with standard values to identify the components.

When the components are colourless, a locating agent is used to visualise them. Ninhydrin spray gives a purple colour to colourless amino acids. Some colourless substances glow when irradiated with ultraviolet light, while some turn brown with iodine vapours.

Q6. Describe the main components of gas chromatography.

Component	Description / function
Carrier gas (mobile phase)	An inert gas such as helium or nitrogen; carries the vapours of the sample through the column.
Injector port	The sample is injected here and quickly vaporised by an oven.
Column (with stationary phase)	A long coiled column packed with tiny silica (or alumina) particles of large surface area, coated with a non-volatile, long-chain, non-polar hydrocarbon liquid of high boiling point; separation occurs here.

Component	Description / function
Oven	Keeps the injector and column hot so that the sample stays as vapour.
Detector	Identifies the separated substances as they leave the column and records the retention time of each.
Chromatogram (recorder)	The graph on which the substances are recorded as peaks; peak areas give the proportions.



Main components of a gas chromatograph

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 28.1

a) A mixture contains butanone and pentane. How can they be separated by using thin layer chromatography using aluminium oxide as stationary phase and benzene as solvent?

A spot of the mixture is placed on the baseline of the alumina plate and the plate is dipped in benzene (non-polar solvent) with the baseline above the solvent level. Butanone is polar (C=O group), so it is adsorbed more strongly on the polar aluminium oxide and travels slowly. Pentane is non-polar, is more soluble in benzene and moves further up the plate. The two therefore separate into different spots: pentane with the larger R_f value and butanone with the smaller one.

b) A mixture of two substances X and Y is analysed by thin-layer chromatography. The R_f value of substance X is larger than that of substance Y. Suggest why substance X has a larger R_f value.

X is less polar than Y. It is adsorbed less strongly on the polar stationary phase and is more soluble in the solvent (mobile phase), so it travels further up the plate.

c) A separate thin layer chromatography experiment was done using the hair dyes A, B and C. The chromatogram obtained is shown. State three conclusions about the hair dyes A, B and C which can be deduced from the chromatogram. [A gives one low spot, B gives one high spot, C gives two spots at the same heights as those of A and B.]

1. Dye A is a single substance (one spot), and so is dye B.
2. Dye C is a mixture of two substances (two spots).
3. Dye C contains both A and B, because its two spots are at the same heights (same R_f values) as the spots of A and B. (A and B are different substances.)

Quick Check 28.2

An impure sample of lactic acid, $CH_3CH(OH)COOH$ contains butanone as the only contamination. The mixture is analysed using gas chromatography. Butanone is found to have a longer retention time than lactic acid. i) Suggest suitable substances that could be used as the mobile and stationary phases. ii) Describe how the percentage composition of the mixture can be determined from the gas chromatogram.

i) Mobile phase: an inert carrier gas such as helium or nitrogen. Stationary phase: a non-volatile, long-chain, non-polar hydrocarbon liquid of high boiling point on silica (or alumina) particles. (Butanone is less polar than lactic acid, so it interacts more with this non-polar liquid and takes longer to pass through.)

ii) Identify the two peaks from their retention times. Measure the area of each peak (½ × base × height). Then:

$$\%\ \text{of lactic acid} = (\text{peak area of lactic acid} \div \text{sum of both peak areas}) \times 100$$

$$\%\ \text{of butanone} = (\text{peak area of butanone} \div \text{sum of both peak areas}) \times 100$$

studysapiens.com